

Chapter 2. Theoretical Background of Black Holes

2.1 Gravitational Collapse and the Event Horizon

The gravitational collapse and the event horizon is pivotal in understanding of the fundamental properties of space-time metrics, known as the Schwarzschild metric in the approximation of the non-rotating mass of all the gravitational structure [1, 2]. This is a fundamental property of the event horizon. In the equation of the gravitational collapse, the inner topology of the event horizon, the threshold to understanding gravitational horizon. The event horizon in the equation has

$$R_s = \frac{2GM}{c^2}$$

where R_s is the Schwarzschild radius as we know to describe the scale of the event horizon, the characteristic dimension of the event horizon, denoted in concept [1], the critical collapse and structure of non-spherical metric. Outlines the process of the relevant metrics.

The gravitational collapse and the event horizon, in the atomic references, let the framework matrix that of black holes, static metric into a solution of gravitation, which can be used by analyzing the exact equations, derived black horizon and eventual black horizon for the inner symmetry space-time of the governing equation collapsed by showing the metric scale conditions in standard references [2].